System Data Files

System Data Files

In Linux, system data files are required for normal operation such as user account authentication and password storage.

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Password File and Shadow File

- In Linux, `/etc/passwd` and `/etc/shadow` are system files that are used for user authentication and password storage.
- `/etc/passwd` aims at user account details while `/etc/shadow` aims at user's password details such as hashed/encrypted password and password expiration date.
- Password file is world accessible. shadow file can only be accessed by root account.
- Specifically, `/etc/passwd` stores basic user account information, including the user's username, user ID (UID), group ID (GID), home directory, and login shell.
- For example, with command `cat /etc/passwd`, for each user, it shows:

  ```
  jmiller5:x:1030:1030:/home/jmiller5:/bin/sh
  <username>:<encrypted passwd>:<user ID>:<group ID>:<home directory>:<shell>
  ```

  The number of fields depending on the system. Some systems have comment and other fields.
- `/etc/passwd` used every time a user log in to the Linux system.

Password File Structure

```
#include <pwd.h>
struct passwd {
   char *pw_name; /*User's login name*/
   char *pw_passwd; /* Encrypted password, not available*/
   uid_t pw_uid; /*Numerical user ID*/
   gid_t pw_gid; /* Numerical Group ID */
   time_t pw_change; /* Passwd changed time, not available*/
   char *pw_class; /* user access class, not available*/
   char *pw_gecos; /* user's full name , and other info.*/
   char *pw_dir; /* user's login directory */
   char *pw_shell; /* user's login shell */
   time_t pw_expire; /* passwd expiration time */
};
```

- Linux define two functions to get entries from the password file.
- The function `getpwnam()` is used by the login program, when a user enter his/her login name.
- The function `getpwuid()` is used by the ls program.

```
#include <sys/types.h>
#include <pwd.h>
struct passwd *getpwnam(const char *name);
struct passwd *getpwuid(uid_t uid);
```

User Authentication Example

```
/*printpasswd.c print out file's infomations*/
#include <sys/types.h>
#include <sys/stat.h>
#include <pwd.h>
#include <stdio.h>
#include <stdlib.h>
void err_sys(char *str) {
  printf ("%s\n",str);
  exit (1);
}
int main(int argc, char *argv[]) {
  struct stat statbuf;
  struct passwd *pwd;
  if (argc != 2)
    err_sys (" argument number error ");
  if (stat (argv[1], &statbuf) < 0) /*get a usefule data from i-node */
    err_sys("stat error for foo");
  /*need pass user id for getting information from passwd*/
  if ((pwd = getpwuid(statbuf.st_uid)) != NULL) {
    printf("Owner's login name: %s \n", pwd->pw_name);
    printf("Owner's passwd: %s \n", pwd->pw_passwd);
    printf("Owner's Numerical user ID: %d \n", pwd->pw_uid);
    printf("Owner's Numerical group ID: %d \n", pwd->pw_gid);
    printf("Owner's login directory: %s \n", pwd->pw_dir);
    printf("Owner's login shell: %s \n", pwd->pw_shell);
    printf("Owner's full Name: %s", pwd->pw_gecos);
}
  else
    printf("%d", statbuf.st_uid);
  exit (0);
}
```

Password Security and Encryption

- Most Unix like systems, use a one-way encryption algorithm, called DES (Data Encryption Standard) to encrypt your passwords, so that it cannot be converted back to plain text.
- This encrypted password is then stored in `/etc/passwd` or in `/etc/shadow`.
- When you login, the password you type is encrypted again and compared with the entry in the file that stores your passwords.
- If they match, the verification is success.
- To make it harder to obtain the encrypted password, Linux store the encrypted password in `/etc/shadow`, only accessible by super user.
- This shadow file can be accessed by few system calls such as `login()` for login and `passwd()` for change password.

System Data Files - Group file

```
#include <grp.h>
  struct group {
  char *gr_name; /* Group name */
  char *gr_passwd; /* Encrypted password. */
  gid_t gr_gid; /* Group ID. */
  char **gr_mem; /* List of group members */
};
```

- The group file is located in `/etc/group`.
- It contains each group related information.
- Linux define two functions to get entries from the group file.

```
#include <sys/types.h>
#include <grp.h>
struct group *getgrgid(uid_t uid);
struct group *getgrnam(const char *name);
```

User Authentication Example

```
/*printgroup.c print out file's group infomation*/
#include <sys/types.h>
#include <sys/stat.h>
#include <grp.h>
#include <stdio.h>
#include <stdlib.h>
void err_sys(char *str) {
  printf ("%s\n",str);
  exit (1);
}
int main(int argc, char *argv[]) {
  struct stat statbuf;
  struct group *gr;
  int i=0;
  if (argc != 2)
    err_sys (" argument number error ");
  if (stat (argv[1], &statbuf) < 0) /*get stat data from i-node */
    err_sys("stat error for foo");
  if ((gr = getgrgid(statbuf.st_gid)) != NULL) {
    printf("Group name: %s\n", gr->gr_name);
    printf("Group number: %d\n", gr->gr_gid);
    while (gr->gr_mem[i]!= NULL){
       printf("%s \n",gr->gr_mem[i] );
       i++;
     }
  }
  else
    printf("%d", statbuf.st_gid);
  exit (0);
}
```

System Data Files – Login/logout Account

Linux provide a few system data files regarding login and logout.

- `/var/log/btmp` keeps track of failed login attempts.
- `/var/run/utmp` keeps track of all the users currently logged in.
- `/var/log/wtmp` keeps track of historical record of utmp data.
- On login, a record of the `utmp` structure is written to the `utmp` file by login program (in binary format).
- On logout, the entry in the `utmp` file was erased by `init` program and a new entry is appended to the `wtmp` file (in binary format).
- The `last` program read the `utmp` file and prints contents in a readable form.

```
struct utmp {
  char ut_line[8]; /* tty name (name of terminal on standard input) */
  char ut_name[8]; /* user name */
  long ut_time; /* time on */
  ...
};

Example command:
last -f /var/log/wtmp
```

Other System Data Files

- Hosts
  - Data file:`/etc/hosts`
  - Header: `<netdb.h>`
  - Structure: `hostent`
  - Functions: `gethostbyname`, `gethostbyaddr`
- Networks
  - Data file: `/etc/networks`
  - Header: `<netdb.h>`
  - Structure: netent
  - Functions: `getnetbyname`, `getnetbyaddr`
- Protocols
  - Data file: `/etc/protocols`
  - Header: `<netdb.h>`
  - Structure: protoent
  - Functions: `getprotobyname`, `getprotobyaddr`
- Services:
  - Data file: `/etc/services`
  - Header: `<netdb.h>`
  - Structure: servent
  - Functions: `getservbyname`, `getservbyaddr`